B. Deck of Cards

time limit per test: 2 seconds
memory limit per test: 256 megabytes

Monocarp has a deck of cards numbered from 1 to n. Initially, the cards are arranged from smallest to largest, with 1 on top and n at the bottom.

Monocarp performed k actions on the deck. Each action was one of three types:

- remove the top card;
- remove the bottom card;
- remove either the top or bottom card.

Your task is to determine the fate of each card: whether it remains in the deck, has been removed, or might be both.

Input

The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases.

The first line of each test case contains two integers n and k ($1 \le k \le n \le 2 \cdot 10^5$).

The second line contains a string s of length k, consisting of characters `0`, `1`, and/or {2}. This string describes Monocarp's actions. If the i-th character is `0`, Monocarp removes the top card on the i-th action. If it's `1`, he removes the bottom card. If it's `2`, either the top or bottom card can be removed.

Additional constraint on the input: the sum of n over all test cases doesn't exceed $2 \cdot 10^5$.

Output

For each test case, print a string consisting of n characters. The i-th character should be `+` (plus sign) if the i-th card is still in the deck, `-` (minus sign) if it has been removed, or `?` (question mark) if its state is unknown.

Example

input Copy

```
4
4 2
01
3 2
22
1 1
2
7 5
01201
```

output Copy

```
-++-
???
-
--?+?--
```

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